
ZIPPO PLATFORM ARCHITECTURE & SPECIFICATION

Spec-Driven Development (SDD) & Test-Driven Development (TDD) Guide

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* All external HTTP requests arrive at port 80 of the shared ALB and receive an immediate '3

* Path patterns evaluate against listener rules in order of priority:

* '/api/*' (Priority 10) forwards to 'zippo-be' target group.

* '/' (Priority 100) forwards to 'zippo-fe' target group.

* In-process sidecars (e.g., Redis container in 'zippo-be') communicate via 'localhost:<port'

* Building and caching base runner Docker images (e.g., node runner, terraform runner).

* Defining the container listening port ('PORT: 3000').

* Defining the health check endpoint ('HEALTH: "/api/v1/health"').

* Defining resource sizing ('CPU: 256', 'MEMORY: 1024', 'DESIRED_COUNT: 1').

* Declaring required sidecar containers (e.g. 'redis-alpine').

* Enabling AWS SES access ('SES: true') if the service sends transactional email.

* Specifying S3 bucket access ('S3_BUCKET: "zippo-files"') if the service handles file uploa

* Maintaining a secure, multi-stage 'Dockerfile'.

* Executing unit, integration, and lint tests in the CI workflow.

* Pushing built images to ECR tagged with '<commit-sha>-<timestamp>'.

* Updating the SSM release tag parameter ('/ci/<service-name>/image_tag').

* Executing 'terraform apply' on the app's 'deploy/<env>' directory.

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- **Runner:** Ephemeral AWS CodeBuild container ('codebuild-ZIPPO-<APP>-<run_id>').

• **Actions:**

* Dependency installation with cache restoration ('~/npm').

* Static code analysis & linting ('npm run lint').

* Automated unit & integration tests ('npm run test:ci').

• **Actions:**

* Resolve AWS Account ID dynamically via 'aws sts get-caller-identity'.

* Compute release tag: 'IMAGE_TAG="\${SHORT_SHA}-\${DATE_TAG}"' (e.g. '41d0162-15-09-26-12-00')

* Authenticate Docker daemon against Amazon ECR:

""bash

* Build container using root 'Dockerfile'.

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- ****SSM Parameter Store Key:**** '/ci/<service-name>/image_tag'

• ****Command:****

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```
aws ssm put-parameter --name "/ci/${SERVICE_NAME}/image_tag" --value "${IMAGE_TAG}" --overwri
```


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- **Rationale:** Decouples image production from infrastructure updates. Terraform reads thi

- ****Path:**** 'deploy/<env>/'

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- **Module Resolution:** Configures Git credentials to fetch the private module 'ZIPPO-INFR/

• **Commands:**

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- ECS registers a new revision of the task definition with the new image tag.

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- Fargate starts new tasks, runs health checks against the target group, verifies healthy st

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1. Developer triggers 'workflow_dispatch' with:

* Service ('zippo-be', 'zippo-fe', etc.)

2. Workflow creates the parameter or secret in AWS with value "PLACEHOLDER".

1. ****Service Identifier:**** Kebab-case service name (e.g. 'zippo-certs').

2. **Repository Location:** GitHub organization and repo name (e.g. 'epddp/ZIPPO-CERTS').

4. **Standard Question:** What container port does the app listen on? (e.g. '3000', '8080').

5. **Standard Question:** What ALB path pattern should route to this service? (e.g. `"/api/v1/`

* The app deployment cannot proceed until the platform team allocates these IDs.

* Shared cluster 'cheap-ecs' only provides Fargate and Fargate Spot capacity.

* Platform team must configure an EC2 or custom capacity provider.

Each environment maintains:

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- 'main.tf': Instantiates 'module "aws_epam_ecs"' targeting the new environment backend stat

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- 'terraform.tfvars': Declares subnet IDs, ALB security group, ACM domain, and CodeBuild pro

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- 'variables.tf': Inherits standard variables.

Run 'terraform apply' to provision:

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- The environment ECS cluster ('zippo-<env>-ecs' or 'cheap-ecs').

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- The environment ALB with HTTPS listener and ACM certificate.

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- The environment CodeBuild runner projects and webhooks.

For each active microservice ('ZIPPO-BE', 'ZIPPO-FE', etc.):

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- Create 'deploy/<env>/config.yaml' populated with environment-specific URLs, database names

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- Create 'deploy/<env>/main.tf', 'locals.tf', 'providers.tf', 'variables.tf', and 'versions.

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- Populate secrets in AWS Secrets Manager and parameters in AWS SSM.

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- Trigger initial deployment.

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- ****Configuration Drift:**** Missing health checks, incorrect ALB priorities, or improperly fo

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- **Security Regressions:** Missing permissions boundaries, insecure environment variables,

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- **Context Hallucinations:** AI assistants guessing non-existent parameters or altering cri

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- The schema definition is stored in 'zippo-specs/schemas/app-config.schema.json'.

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- Validation is executed locally or in CI via:

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- No Terraform code may be planned or applied if schema validation fails.

â satisfy the assertion. Test passes instantly. â

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- **Mock Providers:** `mock_provider "aws"` mocks all API calls, preventing AWS credential d

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- **Command Mode:** Running with 'command = apply' or 'command = plan' checks HCL logic, con

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- **Instant Feedback:** Tests execute in under 3 seconds.

